

EEG Sleep Classification and Reference-Based Sleep-Health Profiling

Existing evidential Transformer with targeted N1/N2 spectral refinement

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HMC data and EEG input

Partition	Recordings	Stored epochs
Training	16	15,216
Validation	4	3,619
Testing	4	3,109
Total	24	21,944

HMC v1.1: 151 clinical recordings; subset 24. Four channels, 100 Hz, 30 s ($4 \times 3,000$). SN001: 95-minute segment.

Retained Fast SCFormer-U architecture

- | $4 \times 3,000$ EEG $\rightarrow$ Conv1D 64 (kernel 7, stride 4) $\rightarrow$ Conv1D 128 (kernel 5, stride 4)
- | Self-attention: 4 heads, key dimension 16; residual normalization and feed-forward layers
- | Bidirectional LSTM: 64 units per direction; pooling and dense evidence head
- | Five softplus evidence outputs; 226,309 total parameters. Attention and recurrence operate within the epoch.

Nine EEG features for N1/N2 refinement

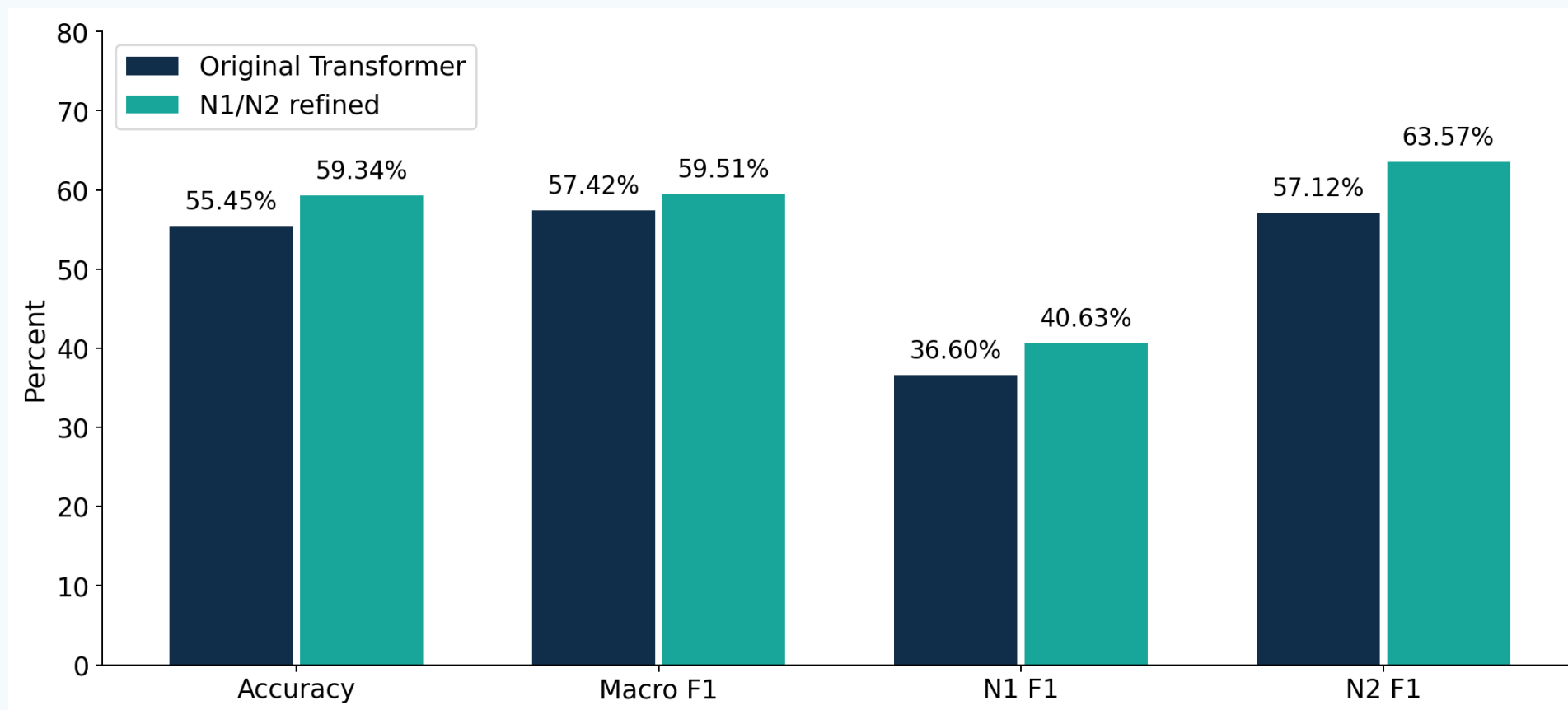
Feature group	Features
Relative band power	Delta, Theta, Alpha, Sigma, Beta
Ratios to Delta	Sigma/Delta, Theta/Delta, Alpha/Delta, Beta/Delta
Frequency bands (Hz)	0.5–4; 4–8; 8–12; 12–16; 16–30
Candidate gate	Transformer predicts N1 or N2

These are project feature boundaries, not AASM staging definitions. Sigma power does not measure spindle density.

Training and inference gates

- Fit the refiner only on training epochs where the true stage AND the Transformer prediction are N1/N2.
- Training subset: 1,278 N1 and 4,588 N2 epochs; 5,866 total.
- StandardScaler → LogisticRegression(class_weight="balanced", max_iter=2000, random_state=42).
- At inference, gate by the Transformer prediction only. Ground truth is never used to route test epochs.

Effect of targeted N1/N2 refinement



Accuracy +3.89 pp; macro F1 +2.10 pp using unrounded values. N1 F1 +4.03 pp; N2 F1 +6.45 pp.

Sleep architecture measurements

Metric	Definition
TST	$N1 + N2 + N3 + \text{REM time}$
Sleep efficiency	$100 \times \text{TST} / \text{evaluated recording duration}$
Stage percentages	$100 \times \text{stage sleep time} / \text{TST}$
WASO	Wake time after first sleep to sequence end
Fragmentation	All stage transitions / TST hours

Recording duration is an evaluated-window measure. Contiguous timing is assumed for the notebook NPZ sequences.

Training reference and four-domain profiles

Reference: 16 training recordings; test matrix: ID + 22 numeric fields. Compare 21 fields, excluding duration of the evaluated window.

$z = (\text{value} - \text{mean}) / \text{sample SD}$. Below P10 → LOW; P10–P90 → NORMAL; above P90 → HIGH.

Domains: Duration (TST); Continuity (efficiency, WASO, awakenings, fragmentation); Initiation (SOL); Architecture (N1/N2/N3/REM %).

Any out-of-range metric marks its domain. 0 / 1 / 2 / 3–4 deviated domains → Within Reference / Mild / Moderate / High Deviation.

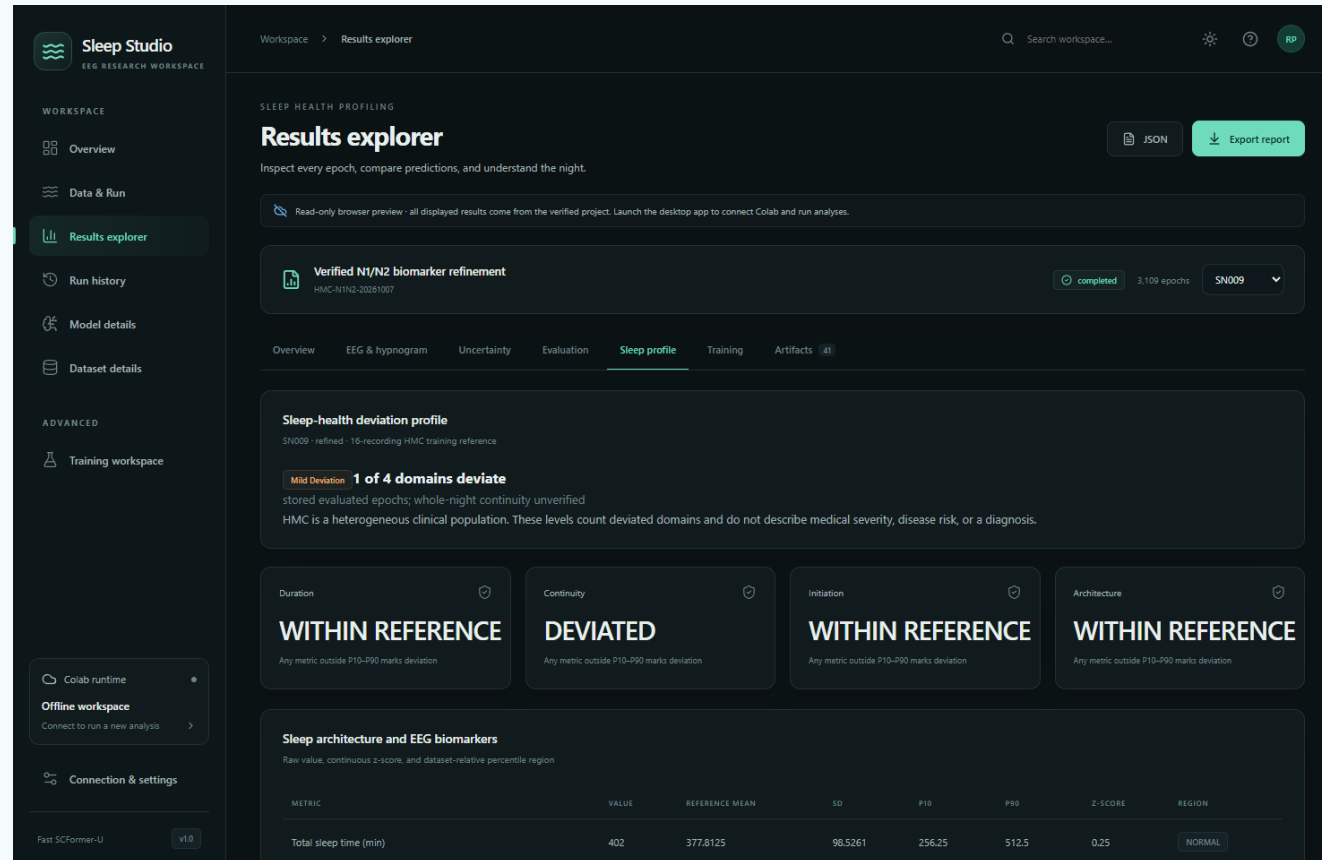
Dataset reference, not healthy norms. EEG deviations are displayed separately; profile levels are descriptive counts.

Test-recording domain profiles

Recording	Duration	Continuity	Initiation	Architecture	Overall
SN001	Deviated	Deviated	Deviated	Within ref.	High Deviation
SN004	Within ref.	Deviated	Deviated	Deviated	High Deviation
SN009	Within ref.	Deviated	Within ref.	Within ref.	Mild Deviation
SN022	Within ref.	Deviated	Within ref.	Deviated	Moderate Deviation

SN001: segment profile. HMC reference deviations are neither disease-risk predictions nor validated severity levels.

Interactive Sleep Studio dashboard



Select SN001/SN004/SN009/SN022. Inspect domains, sleep architecture, nine EEG features, z-scores, P10/P90, and source artifacts.

Complete implemented pipeline

Raw EEG → existing preprocessing → evidential Transformer → targeted N1/N2 spectral refinement

Refined hypnogram → sleep architecture; EEG → spectral biomarkers

Recording feature matrix → 16-recording training reference → z-scores + P10/P90 regions

Four domain statuses → descriptive overall deviation profile → interactive dashboard and report exports

Limitations and unimplemented extensions

- | No explicit spindle detector, dedicated formal SWA metric, or spectral edge frequency.
- | No clinically validated normative ranges, disease prediction, or diagnostic severity classification.
- | No documented repeated-night participant mapping for true longitudinal personalization.
- | Small clinical reference (16 recordings), assumed NPZ continuity, in-sample reference architecture, and a partial SN001 input.